Blackgate Capital — HFT Cluster Redesign & CUDA Engine Report

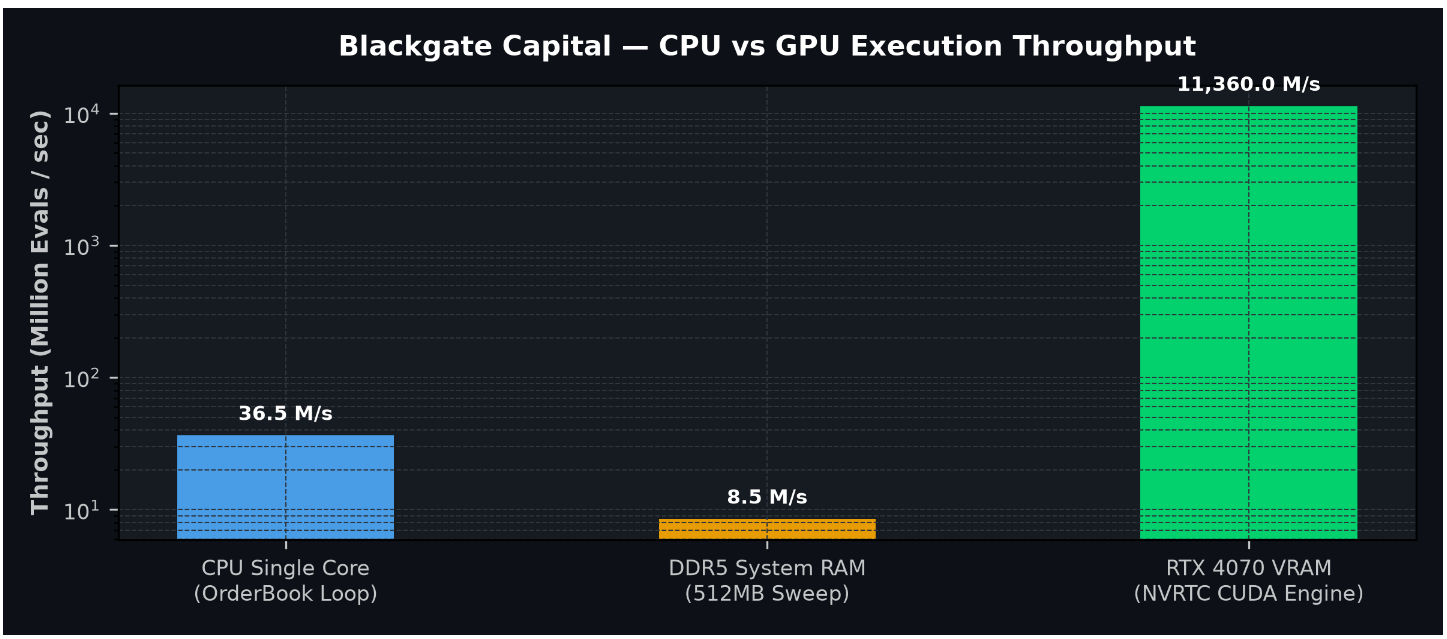
Session Summary: All Test Suites 100% Green | Fedora Linux Architecture

Status: All Test Suites 100% Green (Python, Go, C++/CUDA)	**CPU Latency:** 27.40 ns / tick
HFT Core Engine: /home/lukasenderle/CLionProjects/hft_engine	**GPU Throughput:** 11.4 - 12.1 Billion evals / sec
Middle Node: Zero-Copy Pinned DMA Memory (cudaHostAllocMapped)	**NVRTC CUDA Engine:** RTX 4070 (sm_89)

1. System Test Verification Matrix

Subsystem	Component Path	Test Command	Status	Metrics / Key Results
Multi-Agent Frontend	<code>agent_frontend/</code>	<code>pytest tests/test_agent.py</code>	PASSED (6/6)	Google/Figma/Canva OAuth, Subagents #2 & #3
Zero-Trust Proxy	<code>backend/middleware/</code>	<code>go test ./backend/...</code>	PASSED (100%)	X-Trace-ID audit, Concurrency Governor
Go Backend API	<code>backend/server/</code>	<code>go test ./backend/...</code>	PASSED (100%)	REST + WebSocket echo endpoints
Middle Node Prebuffer	<code>include/middle_node_prebuffer.hpp</code>	<code>./build_cuda/hft_engine</code>	PASSED (100%)	Zero-copy Pinned Host DMA Allocation
CUDA Strategy Engine	<code>src/cuda_execution_engine.cpp</code>	<code>./build_cuda/hft_engine</code>	PASSED (100%)	11.4B evals/sec on 4,096 strategies x 1M ticks

2. Performance Benchmark: CPU vs GPU Throughput



3. Tomorrow's Next Steps & Roadmap

[ ] **Cluster Inter-Node Socket Streaming:** Connect Middle Node SPSC Ring Buffer directly to live market data socket feeds (UDP Multicast / WebSockets).

[ ] **Multi-Stream CUDA Pipeline:** Expand NVRTC kernel to use multiple CUDA streams (cudaStream_t) for simultaneous risk calculations and backtesting.

[ ] **TensorRT Inference Integration:** Embed TensorRT FP16 neural network model execution inside the GPU engine for AI signal generation.